

Deepak Mallampati

AI PLATFORM / MLOPS ENGINEER

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Kent, OH - US-based, F-1 OPT (open to NY/SF hybrid)

PROFESSIONAL SUMMARY

AI Platform / MLOps engineer with 4+ years building production GenAI infrastructure: LLM CI/CD pipelines, evaluation frameworks (RAGAS, BERTScore), model-serving APIs, and inference cost/latency optimization. Shipped 12 FastAPI services on Kubernetes handling 100K+ daily requests with p95 latency cut from 1.5s to 800ms via prompt compression, semantic caching, and model routing. Hands-on with PyTorch, MLflow, Bedrock/SageMaker, Docker, K8s, and QLoRA fine-tuning on H100 - looking to go deeper into open-source inference engines (vLLM, Sglang, TRT-LLM) at LLM-serving scale.

SKILLS

LLM CI/CD & Evaluation • Inference Optimization • Model Serving & Control Plane • MLOps Infrastructure • Kubernetes & Docker • PyTorch & MLflow • Quantization (QLoRA, 4-bit) • Semantic Caching & Model Routing

ML Infrastructure: PyTorch, TensorFlow, MLflow, Weights & Biases, AWS Bedrock/SageMaker, Azure OpenAI/AI Services, GCP Vertex AI, Docker, Kubernetes, Terraform, CI/CD (Jenkins, Azure DevOps)

LLM Engineering: GPT-4o/5, Claude Sonnet, Gemini, LLaMA, Qwen3-72B (QLoRA fine-tune on H100);
LangChain/LangGraph; semantic caching; model routing; prompt compression; KV cache & prefill awareness

Evaluation & Observability: RAGAS, BERTScore, custom domain metrics, latency & accuracy dashboards, regression detection, A/B testing, drift detection

Retrieval / RAG: FAISS, Pinecone, Weaviate; dense + hybrid search; cross-encoder re-ranking; chunking strategies

Languages & APIs: Python (primary), JavaScript/TypeScript, Java, SQL; FastAPI, React; REST, JSON, async patterns

Security & Compliance: HIPAA, SOC 2, PCI DSS; OAuth 2.0, JWT, RBAC; PII masking; prompt-injection mitigation; audit logging

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Established an **LLM evaluation & monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency and accuracy dashboards, surfacing regressions **before release** across multi-step agentic workflows.
- Optimized inference cost and latency via **prompt compression, semantic caching, and model routing**, sustaining SLA targets while reducing per-query overhead.
- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents - reducing token consumption by **42%** while sustaining task accuracy.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination.
- Built fault-tolerant **automation infrastructure** featuring scheduled batch orchestration and exponential-backoff retry logic, improving pipeline reliability and reducing failed production runs by **60%**.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from **1.5s to 800ms** and accelerating feature integration by **40%** - handling **100,000+ requests/day** across 20+ internal teams.
- Developed and enhanced **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ agents**.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Engineered safeguards and content controls for policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100, conditioning on a six-stage narrative schema for controllable long-form generation.
- Built a privacy-preserving ML pipeline (k-anonymity, l-diversity, differential privacy) - reducing record-linkage re-identification risk from **3.38% to 0.32%** while retaining **99% of baseline predictive performance**.
- Authored an IEEE-format conference paper documenting methodology, experiments, and results.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Developed supervised regression and classification models on operational pipeline-storage data; equipment-failure prediction, anomaly detection, capacity forecasting.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification on domain-specific financial datasets, achieving **89% F1-score**.

PROJECTS

MangaLens [Chrome Web Store](#)

Shipped Chrome extension delivering real-time AI translation of manga and webtoons across **29+ sites**. Turned a multi-step manual translation chore into a single inline action.

JavaScript/TypeScript, LLM APIs, browser extension architecture

career-ops [Open Source](#)

Autonomous AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - multi-agent orchestration with structured evaluation.

Python, Node.js, LangGraph, Playwright, structured-output LLMs

Story Director LLM

Fine-tuned (QLoRA) language model that generates structured short-form video narratives from a single prompt, paired with an automated rendering pipeline.

PyTorch, QLoRA, Qwen3-27B, H100 GPU

RESEARCH

Privacy-Preserving Clinical ML Pipeline

Framework letting researchers analyze sensitive patient data without exposing identities - combining k-anonymity, l-diversity, and differential privacy. Quantified the privacy-utility trade-off across six anonymization configurations.

scikit-learn, differential privacy (Laplace mechanism), Random Forest

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025

GPA: 3.70 / 4.0

WORK AUTHORIZATION

F-1 OPT - authorized to work in the United States now; future sponsorship required.